Milton Tran

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Education

University of California, Riverside — Riverside, CA
Undergraduate Student in Mechanical Engineering — *Expected Graduation: June 2030*

Engineering Experience

FRC Team 9470 — Fremont, CA
Mechanical Design Lead — *June 2024 – May 2026*

- Led mechanical design and integration, defining robot architecture and coordinating subsystem layouts to support overall performance and serviceability.
- Designed and optimized a shooting mechanism capable of emptying a hopper containing 60+ game pieces in under 3 seconds.
- Coordinated pit operations, maintenance, and mechanical troubleshooting during competitions to restore robot functionality under time constraints.
- Collaborated with FRC Teams 1678 and 254 at district and world championship events to exchange engineering ideas and improve robot performance.

Highlander Racing — Riverside, CA
Suspension Subteam — *September 2026 – Present*

- Design suspension and steering components in SolidWorks, accounting for packaging constraints, geometry, and mechanical integration.
- Apply FEA to evaluate component stresses and Analyze suspension geometry and mechanical behavior to support vehicle performance and reliability.

Highlander Combat Robotics — Riverside, CA
Mechanical Subteam — *October 2026 – Present*

- Design mechanical components and assemblies in Onshape for a plastic combat robot, considering fit, function, and manufacturability.
- Develop lightweight mechanical designs while balancing structural requirements, packaging constraints, and ease of assembly.

Projects

More on portfolio

Custom CNC Router | *Mechanical Design, CAD, GRBL*

- Designed and assembled a custom CNC router using 2020 aluminum extrusions, linear rails, lead screws, and 3D-printed components.
- Integrated NEMA 17 stepper motors with an Arduino Uno running GRBL and used Universal G-code Sender (UGS) to control machine movement.

Two-Axis Gimbal and Cycloidal Drives | *CAD, FEA, Inverse Kinematics*

- Designed and iteratively tested cycloidal drives, refining the output mechanism and gear ratio while packaging the final design within a 70 mm diameter.
- Derived inverse-kinematics equations to calculate motor angles for target end-effector coordinates in a planar robot with two interconnected arms.

Awards

FIRST World Championship: 2026 Archimedes Division Champion
FIRST District Championships: 2026 Half Moon Bay and Silicon Valley District Champion
Engineering Awards: FIRST Excellence in Engineering and Industrial Design Awards

Technical Skills

CAD and Design Software: Onshape, SolidWorks, Fusion 360, OrcaSlicer
Manufacturing and Fabrication: FDM and SLA 3D Printing, CNC Machining, Assembly
Electronics: Arduino Uno, Universal G-code Sender, Wiring
Programming and Tools: Python, Java, VS Code